26900251 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCFSC300 - FORENSIC BALLISTICS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1. What is a double-action firearm? [U] / [CO1]
2. What are shock waves associated with a projectile? [U] / [CO3]
3. What is meant by matching crime bullets with test bullets? [U] / [CO4]
4. What is a primer composition? [U] / [CO3]
5. Mention any two factors affecting pressure inside a firearm barrel. [U] / [CO3]
6. What is the Walker test? [U] / [CO5]
7. Explain the forensic significance of bullet striations. [U] / [CO4]
8. A bullet loses its gyroscopic stability during flight. What effect would this have on its motion. [A] / [CO5]
9. What are the characteristics of a contact shot? [U] / [CO4]

10. What is the scope of forensic ballistics in criminal investigation? [U] / [CO1]

11. A bullet strikes a hard surface at an angle and changes its direction. Identify and explain the phenomenon involved. [A] / [CO5]

12. Differentiate between muzzle flash and muzzle blast. [U] / [CO3]

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **5** questions

Each question carries **4** marks

13. Explain the role of propellants in ammunition and mention their important characteristics. [K] / [CO1]

14. Explain the role of the National Integrated Ballistic Information Network (NIBIN) in linking firearm-related incidents. [U] / [CO3]

15. Describe the important identifying features of a 12-bore firearm and a .315 firearm. [U] / [CO2]

16. Describe how the range of firing for a shotgun can be determined. [A] / [CO4]

17. Describe the Price test and its significance in firearm examination. [U] / [CO5]

18. Define intermediate ballistics and explain its importance in the study of firearm projectile motion. [U] / [CO2, CO3]

19. Describe the magazine and firing mechanism of a firearm and explain their functions [K] / [CO1]

[4x5 = 20]

Part C

Essay Type Questions

Answer any **1** questions

Each question carries **10** marks

20. Explain the significance of GSR composition, positioning and collection in establishing a connection between a person and a firearm incident. [A] / [CO5]

21. Discuss the forensic examination of improvised, country-made, and imitative firearms and the challenges associated with their identification. [U] / [CO1]

[10x1 = 10]